Assignment 5B: ELO Calculations

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For this assignment, I will continue working with the chess tournament data from Project 1. I will use each player's pre-tournament rating and the ratings of their individual opponents to calculate their expected tournament score using an ELO expected-score formula.

I will first load the original tournament data into R and extract each player's opponents and their pre-tournament ratings. I will then calculate the expected score for each game based on the difference between the two players' ratings. I will add these values together to calculate each player's total expected score.

Next, I will compare each player's expected score with their actual score and calculate the difference between the two. A positive difference will indicate that a player performed better than expected, while a negative difference will indicate that a player performed worse than expected.

Finally, I will identify the five players who most overperformed and the five players who most underperformed relative to their expected scores. I will present the results in tables and cite the source of the ELO formula used in my calculations.